

Hybrid Neural Engine (MLP + Random Forest)

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1 Introduction

The Hybrid Model represents the core inferential layer for directional 0-DTE options trading operations. It operates by generating probability outputs regarding intra-day asset price action (SPX) relative to short-term strike barriers. The architecture utilizes a dual-engine ensemble combining a regularized Multilayer Perceptron (MLP) within PyTorch, and a complementary Random Forest (RF) classifier acting as a non-linear interaction filter.

2 Mathematical Formulation of the Target Variable

The classification target $\mathcal{Y} \in \{0, 1, 2\}$ is defined as a non-overlapping categorical state indicating optimal market execution directions:

$$y_i = \begin{cases} 1 & \text{if } R_{30m}^f \geq +0.06\% \quad (\text{LONG}) \\ 2 & \text{if } R_{30m}^f \leq -0.06\% \quad (\text{SHORT}) \\ 0 & \text{otherwise} \quad (\text{HOLD}) \end{cases}$$

where R_{30m}^f is the 30-minute forward geometric return of the underlying asset. The threshold of $\pm 0.06\%$ accounts for expected slippage, commissions, and standard bid-ask spreads on high-gamma SPX contracts.

3 Architectural Design

3.1 Multilayer Perceptron (MLP) Specifications

The MLP maps the standardized continuous $\mathcal{X} \in \mathbb{R}^{92}$ dimensional matrix into a continuous $\mathbb{R}^3$ logit representation. Let $\mathbf{x}^{(0)}$ be the initial feature state. The forward pass is defined as:

$$\begin{aligned} \mathbf{z}^{(l)} &= \mathbf{W}^{(l)} \mathbf{x}^{(l-1)} + \mathbf{b}^{(l)} \\ \mathbf{x}^{(l)} &= \text{Mish}(\text{LayerNorm}(\mathbf{z}^{(l)})) \odot \text{Dropout}(p) \end{aligned}$$

for layers $l \in \{1, 2, 3\}$, sequentially reducing dimensionality: $92 \rightarrow 256 \rightarrow 128 \rightarrow 64$.

The final probability distribution is extracted via the Softmax normalization operator:

$$P(\hat{y} = k \mid \mathbf{x}) = \frac{\exp(\mathbf{z}_{\text{out}}^{(k)})}{\sum_{j=0}^2 \exp(\mathbf{z}_{\text{out}}^{(j)})}$$

3.2 Random Forest Class Weighting & Voting

Gradient-boosted decision trees inherently manage categorical imbalances without aggressive undersampling. The secondary model partitions the feature space $\mathcal{X}$ into discontinuous hyper-rectangles R_m :

$$H_{\text{RF}}(\mathbf{x}) = \arg \max_{c \in \{0, 1, 2\}} \sum_{t=1}^{N_{\text{trees}}} \mathbf{1}_{\{h_t(\mathbf{x})=c\}}$$

The system demands strict unanimity. Execution is triggered only if:

$$P(\hat{y}_{MLP} = c) > \tau \quad \wedge \quad H_{RF}(\mathbf{x}) = c$$

where τ is the mathematically determined operational threshold natively optimized over out-of-sample periods (typically $\tau \approx 0.60$).

4 Training Regimen and Overfitting Mitigation

4.1 Walk-Forward Validation

The model entirely dismisses traditional randomized cross-validation (K-Fold) to prevent strict temporal leakage. Training epochs advance via a staggered sliding window manifold:

- **Training Horizon:** $t - 180 \text{ Days} \rightarrow t$.
- **Validation Horizon:** $t \rightarrow t + 30 \text{ Days}$.

4.2 Cross-Entropy Objective and Focal Penalties

The minimization parameter space applies a weighted Negative Log-Likelihood utilizing dynamically computed inverse class frequencies ω_k :

$$\mathcal{L}(\theta) = - \sum_{i=1}^B \omega_{y_i} \log P(\hat{y}_i \mid \mathbf{x}_i; \theta) + \lambda \|\mathbf{W}\|_2^2$$

where L_2 regularization ($\lambda = 10^{-4}$) is applied across all dense linear constraints to ensure continuous geometric generalization outside of historical state bounds.